Validation and Verification Report

Assessment of the NuLife GreenTech: Ontario Avenue Bio-Oil Sequestration Project

Version Control

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Report for:

Isometric and NuLife

https://isometric.com/
https://www.nulifegreentech.com/

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Formal validation of project

Normec Verifavia has been contracted by Isometric on behalf of NuLife to conduct a validation of the NuLife bio-oil carbon removal project against the requirements specified in the Isometric Bio-Oil Geological Storage Protocol, including the Feedstock Characterization, Energy Use Accounting, Transportation Accounting, and Storage Accounting modules. The objective is to confirm that the carbon removals claimed are within scope, real, quantifiable, additional, verifiable, counted once, under clear ownership, and that the project complies with monitoring and reporting requirements.

Normec Verifavia conducted a validation to provide assurance to Isometric that the life-cycle analysis (LCA) model, sampling procedures, and reporting practices for the period under review are free of non-conformances and material misstatements. Upon review of the submission materials, Normec Verifavia conducted a risk assessment to determine the sampling and validation methodology. The verifiers reviewed the supporting documentation according to the sampling and validation plans.

1. Introduction

1.1 Project Presentation

NuLife GreenTech Inc. is a Canadian carbon removal company based in Saskatoon, Saskatchewan. The company develops and operates a facility that use Hydrothermal Liquefaction (HTL) technology to convert waste biomass into a stable, carbon-rich bio-oil. Instead of being landfilled or left to decompose, waste streams such as dairy processing sludge, oat residues, agricultural by-products, and fats, oils and greases are diverted and transformed into bio-oil.

This bio-oil is then permanently stored in deep underground salt caverns through a regulated injection process. These caverns, formed millions of years ago and historically used for natural gas and liquid fuel storage, provide a secure geological environment where carbon is locked away for more than 1,000 years.

By combining waste management with durable carbon storage, NuLife creates a net-negative climate solution:

- Preventing emissions from waste decomposition or land application,
- Transforming biomass carbon into a stable form, and
- Permanently isolating that carbon from the atmosphere.

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In line with the requirements in Isometric's Standard, Protocol and relevant Modules, NuLife quantifies, monitors, and verifies the carbon removed by its operations. The resulting high-quality Carbon Dioxide Removal (CDR) credits help organizations and governments meet their climate commitments while accelerating the development of innovative carbon removal infrastructure in Canada.

1.2 Scope

Normec Verifavia has been appointed by Isometric to provide independent assurance on the quality of data and methodology used to assess performance against key indicators under the Isometric standard, including:

- Measurement and reporting of carbon removal volumes associated with the Hydrothermal Liquefaction (HTL) process and subsequent bio-oil storage.
- Monitoring of biomass feedstock sourcing and treatment volumes at the Saskatoon facility.
- Verification of data supporting NuLife's capacity expansion and operational scaling.

To confirm this information and its coherence with Isometric's requirements, an on-site visit was conducted on the 20th of August 2025.

1.3 Audit Methodology

The assurance process included interviews with NuLife staff responsible for monitoring and measuring plant performance to gain insight into the data collection process and to evaluate the quality of the data used for reporting. In addition, the audit team conducted both on-site validation and verification activities at the Saskatoon facility and desk-based reviews of supporting documentation, records, and calculations.

The audit methodology was designed with the following key objectives:

- To confirm that project activities and monitoring practices align with the requirements of the Isometric Bio-Oil Geological Storage Protocol (V1.1).
- To assess whether carbon removal volumes are real, quantifiable, additional, and verifiable.
- To determine whether reported data is free from material misstatement and non-conformance.
- To evaluate that monitoring and reporting systems provide consistent, transparent, and traceable records.
- A validation plan was shared with NuLife in advance (see Appendix 3), outlining the criteria, scope, and sampling strategy. This plan specified which data streams, calculations, and operational records would be reviewed through interviews and site inspection, and which would be validated through desk review.

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- Site visit activities included: observation of HTL processing operations, review of sampling and testing procedures, confirmation of feedstock receiving and weighing processes, inspection of storage tanks, and examination of safety and monitoring protocols.
- Desk review activities included: assessment of life-cycle analysis (LCA) models, validation of emission and removal calculations, review of Renewable Energy Certificate (REC) purchases, confirmation of biomass sourcing documentation, transport and injection records, and brine quality monitoring data.
- Risks identified in the validation methodology—such as reliance on third-party laboratories, ownership of carbon attributes, and uncertainty in feedstock carbon content—were mitigated by adopting conservative assumptions, reviewing multiple data sources, and requiring supporting documentation (e.g., supplier attestations, injection site agreements, calibration certificates).
- Normec Verifavia conducted the independent assurance audit based on the following criteria:
    - ISO/IEC 17029:2019 – Conformity assessment – General principles and requirements for validation and verification bodies.
    - ISO 14065:2020 – General principles and requirements for bodies validating and verifying environmental information.
- The GHG Protocol – Corporate Accounting and Reporting Standard.
- Normec Verifavia AMSM (Version 9.9, July 2025).
- Isometric Standard v1.7.3 (July 2025).
- Isometric Bio-Oil Geological Storage Protocol (V1.1).
- Isometric Modules:
    - Biomass Feedstock Accounting (V1.2)
    - Bio-oil Storage in Salt Caverns (V1.1)
    - Energy Use Accounting (V1.2)
    - Transportation Emissions Accounting (V1.1)
    - Embodied Emissions Accounting Module (V1.0)

1.4 Responsibilities

NuLife is solely responsible for preparing and reporting its carbon related data, including tracking files, calculation models, primary data, and any supporting information or assessments. Additionally, NuLife is responsible for defining, establishing, and maintaining appropriate performance management and internal control systems from which the reported information is derived.

In accordance with the assurance agreement, our responsibility is to form an independent opinion based on an examination of the information and data presented in the NuLife reporting system, Isometric's submissions, and other supporting documents, and to report that opinion to NuLife. Additionally, we report if, in our opinion:

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- The presented data contains or may contain misstatements (omissions, misrepresentations, or errors) or non-conformities.
- The verification team has not received all necessary information and explanations required to conduct its examination.
- Improvements can be made to NuLife's performance in monitoring and reporting.

2. Technology, Processes, and Inputs

2.1 Baseline scenario

NuLife's documentation describes the baseline scenario as the *business-as-usual management of waste biomass in the Saskatoon region*. In the absence of the project, the identified feedstocks (e.g., dairy DAF sludge, oat hulls, from [redacted], bran, agricultural residues, fats, oils and greases) would typically be disposed of through land application, landfilling, or wastewater treatment. Under these counterfactual pathways, the carbon stored in the biomass would be released back into the atmosphere as biogenic CO_2 or methane within a relatively short timeframe.

Normec Verifavia validated this baseline description through:

- Review of feedstock supplier records and attestations confirming that the materials are treated as waste and have no viable market use as commodities.
- Cross-checking disposal practices in Saskatchewan for similar feedstocks to confirm that landfill, wastewater treatment, or land application is the standard management method.
- Assessment of counterfactual analysis in NuLife's PDD, which excludes any ineligible biomass (i.e., feedstocks that would have remained durably stored or could reasonably be used for energy generation).
- Interviews with NuLife staff and feedstock providers, which confirmed that the project does not divert biomass from existing long-term storage or energy applications.

Based on this review, the baseline scenario was found to be consistent with protocol requirements: the waste biomass would otherwise decompose or be processed in a way that rapidly returns carbon to the atmosphere, and no durable storage of carbon would occur.

2.2 Project boundaries

NuLife's documentation defines the project boundaries to include all activities directly related to the conversion of biomass into bio-oil and its subsequent permanent storage. Specifically, the boundaries cover:

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- Upstream feedstock handling – delivery of waste biomass to the Saskatoon HTL facility, including transport from local suppliers such as [redacted] and [redacted].
- HTL processing operations – conversion of biomass into bio-oil, wastewater, and tailgas at the Saskatoon plant.
- Onsite energy use and consumables – electricity (tracked by facility power meters, with RECs purchased to offset grid emissions), water use, and cleaning chemicals.
- Transport of bio-oil – road shipment by third-party haulers from Saskatoon to [redacted] storage site near [redacted].
- Geological storage – injection of bio-oil into regulated salt caverns, brine displacement and management via a disposal well, and required sampling and monitoring.

Normec Verifavia validated these boundaries through:

- Document review of NuLife's PDD, their submissions to Isometric, Isometric submissions, and supporting diagrams that delineate included and excluded activities.
- Site visit confirmation at the Saskatoon HTL plant, where feedstock receiving, storage, processing, and monitoring systems were observed.
- Inspection of transport documentation (e.g., bills of lading, delivery records) for bio-oil shipments to [redacted].
- Review of regulatory permits for [redacted] caverns and brine disposal well to confirm that injection and storage activities fall within project scope.

Based on this assessment, Normec Verifavia concluded that NuLife's defined project boundaries are consistent with the requirements of the Isometric Bio-Oil Geological Storage Protocol (V1.1) and cover all material sources and sinks relevant to carbon accounting.

2.3 Technology Overview

NuLife GreenTech operates a Hydrothermal Liquefaction (HTL) facility in Saskatoon, Saskatchewan, Canada. HTL is a continuous thermo-chemical process that converts high-moisture organic waste streams into a carbon-rich liquid bio-oil. This bio-oil is then transported to a licensed underground salt cavern near [redacted], where it is permanently sequestered in accordance with the Isometric Bio-Oil Geological Storage in Salt Caverns.

The facility began design in 2022, installed its first processor in 2023, expanded with a second processor in 2024, and is scaling toward a 12-processor commercial module in Q4 2025. The plant is expected to operate through at least 2050.

2.4 Process Flow

1. **Feedstock Sourcing & Delivery**

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a. NuLife sources organic waste biomass within approximately a 25 km radius of Saskatoon.
b. NuLife's feedstock providers are [redacted].
c. Eligible feedstocks include dairy Dissolved Air Flotation (DAF) sludge, grain milling residues, oat hulls, agricultural residues, Fats, Oils and Greases (FOG), and other food manufacturing wastes.
d. Dry biomass is stored onsite; wet biomass is held in 10,000 L storage tanks.

2. **HTL Conversion**
   a. Feedstock is mixed into slurry and pumped into the HTL processor.
   b. The process reforms the carbon in the waste into long-chain hydrocarbons (bio-oil).
   c. By-products include:
      i. Bio-oil – captured for storage.
      ii. Wastewater – recycled into the process or treated/disposed offsite.
      iii. Tailgas – ~99% biogenic CO_2, vented to atmosphere.
      iv. Biochar – emulsified with bio-oil and captured for storage when produced.
3. **Storage & Transport**
   a. Bio-oil is accumulated in external tanks until sufficient volume is available.
   b. A third-party hauler transports the bio-oil to the [redacted] injection site.
4. **Geological Injection**
   a. At the injection site, bio-oil is offloaded, pumped underground, and stored in a salt cavern.
   b. As bio-oil is injected, brine is displaced, monitored, and safely managed according to provincial permitting requirements.
5. **Human capital**
   a. NuLife Greentech has a total of 8 employees. Two of the employees' commute were taken into account for the life-cycle analysis (LCA).

2.5 Inputs and Supporting Materials

- **Biomass Feedstocks:** Organic wastes from food and agricultural sectors (DAF sludge, bran, oat hulls, milling residues, etc.).
- **Energy:** Primarily electricity from the Saskatchewan grid; Renewable Energy Certificates (RECs) are purchased to offset grid intensity.
- **Consumables:** Small amounts of sodium hydroxide and hydrogen peroxide (for cleaning) activated carbon (for gas scrubbing), and municipal water for start-up/cooldown.
- **Transport:** Trucks are used for feedstock delivery, bio-oil transport, wastewater hauling, and sample delivery.

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2.6 Conversion of biomass into bio-oil

The project achieves durable carbon removal by converting waste biomass into bio-oil and permanently storing it underground. Counterfactual analysis confirms that, without this project, the waste biomass would decompose, releasing its carbon back to the atmosphere.

On average, approximately 70% of the incoming biomass feedstock is converted into bio-oil through the Hydrothermal Liquefaction (HTL) process. Of this bio-oil, about 75% of the mass consists of elemental carbon. To calculate the amount of carbon dioxide removed and stored, NuLife applies the following formula:

metric tonnes CO_2 sequestered = 1 metric tonne of bio-oil×0.75(carbon fraction) ×3.67(molecular weight conversion of C to CO_2)

This calculation reflects the stoichiometric conversion of carbon mass to its equivalent carbon dioxide mass, providing the basis for quantifying durable carbon removals under the Isometric protocol.

2.7 Carbon Removal Storage

NuLife quantifies stored carbon based on the mass of bio-oil produced, its measured carbon content, and the stoichiometric conversion of carbon to carbon dioxide. On average, approximately 70% of the incoming biomass feedstock is converted into bio-oil through the HTL process. the conversion of biomass to bio-oil is quantified on a daily basis via continuous mass and energy balance work, however, laboratory analysis shows that roughly 75% of the bio-oil mass consists of elemental carbon. Using the molecular weight conversion factor of 3.67 (C → CO_2), NuLife applies the following calculation:

Total tCO_2e stored = bio-oil (kg) × fraction of elemental carbon content × stoichiometric conversion

For the current crediting period (July 23rd 2025 to September 3rd 2025), the amount of carbon removed is equal to:

Total tCO_2e stored = 21,925 kg x 0.7413 x 3.67 = ***59.600 tCO_2e***

This calculation approach aligns directly with the Isometric Bio-Oil Geological Storage Protocol (V1.1), which requires quantification of carbon removals based on measured carbon mass fractions and conversion to CO_2 equivalents. Bio-oil samples are collected three times per day and tested in duplicate onsite for moisture content. They are also sent to the University of Saskatchewan for independent elemental carbon analysis, including duplicate carbon wt% runs of each sample.

Once produced, the bio-oil is accumulated in external holding tanks and then transported by licensed third-party haulers to [redacted] injection site near [redacted]. At

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the site, the bio-oil is offloaded at an injection pad and pumped underground into deep salt caverns within the Prairie Evaporite Formation. Displaced brine is transferred to a regulated saltwater disposal well in the Lodgepole formation, with weekly onsite sampling and monthly laboratory tests confirming water quality.

Currently, two caverns (Caverns 1 and 4) are active with ~50,000 m³ remaining each, while Cavern 5 (175,000 m³) is permitted and expected to enter use once operational. The caverns were originally constructed for hydrocarbon storage in the 1950s–60s, are naturally pressurized at ~300 psi, and are subject to mechanical integrity testing every five years. No negative pressure events have been reported.

By combining accurate carbon accounting with regulated and geologically secure storage, NuLife ensures that the carbon embedded in its bio-oil is **permanently isolated from the atmosphere for more than 1,000 years**, in line with Isometric's permanence requirements.

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3. Validation and Verification Summary

3.1 Opinion

The validation confirms that NuLife's Saskatoon project uses a Hydrothermal Liquefaction (HTL) system to convert waste biomass into bio-oil for permanent geological storage in regulated salt caverns operated by Fleet Energy.

The processes, inputs, monitoring, and reporting practices are consistent with the Isometric Bio-Oil Geological Storage Protocol (V1.1). Operational procedures such as feedstock input verification, bio-oil sampling, brine testing, and data recording are in place. Safety procedures, such as requiring all visitors to sign a consent form before entering the facility, are also implemented.

3.2 Assurance Methodology

The assurance process included interviews with NuLife staff responsible for monitoring and reporting operational and carbon removal data. The audit team, led by Patricia Pinilla as Lead Auditor and supported by Amit Mehta as Auditor, reviewed NuLife's data collection procedures, tracking systems, and applied methodologies under the Isometric program.

3.3 Independent Statement

We confirm that Normec Verifavia and the verification team are independent of both NuLife and Isometric and have not been involved in preparing feedstock data, carbon accounting calculations, or any documentation provided to or from Isometric, except for this Assurance Statement.

3.4 Assurance Opinion: Supporting Data & Monitoring Methodology

The verification process confirms that NuLife's methodology to monitor progress against its key indicators is fit for purpose, with clear targets and consistent application. The monitoring process is supported by accurate and reliable data, and appropriate quality checks have been applied.

Period	Biomass Slurry (mixed with water) Processed (Liters)	Bio-oil Produced (kg)	Carbon Sequestered (tCO_2e)	Process and project emissions (tCO2e)	Net Carbon Sequestered (tCO_2e)
23.07.2025-03.09.2025	130,230	21,925	59.600	3.862	55.700

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3.5 Validation and Verification Opinion

The objective of this validation and verification was to confirm that NuLife's performance attributes and carbon removal claims are real, quantifiable, and verifiable, and that the project has met the requirements of the Isometric Bio-Oil Geological Storage Protocol (V1.1) and associated modules.

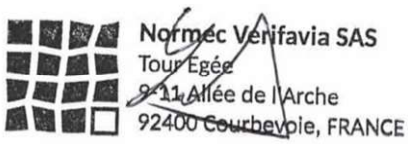
During the auditing period, all findings raised in the issues log were resolved. Normec Verifavia identified 19 clarifications, omissions, or missing requirements related to submitted documents and data. A detailed breakdown of these issues and their resolution is provided in Appendix 1.

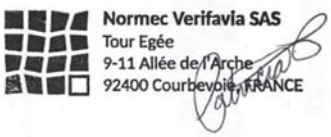
In conclusion:

- NuLife has prepared and submitted its GHG Statement to Isometric free of material misstatement; and
- NuLife has prepared and submitted its GHG Statement to Isometric in full conformance with the requirements of the Isometric Bio-Oil Geological Storage Protocol (V1.1) and associated modules.

The result is a **Positive Validation and Verification Statement**, the basis for which is detailed in this validation and verification report and the accompanying evidence.

Leonard Barkley	Patricia Pinilla
Lead Auditor	Independent Technical Reviewer
Normec Verifavia	Normec Verifavia





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Criteria	Evidence	Protocol / Standard Section	Compliant	Comments
Scope	Project activity is HTL conversion of waste biomass into bio-oil, with permanent storage in [redacted] salt caverns [redacted] Eligible feedstocks include DAF sludge, oat hulls, bran, agricultural residues, FOG, etc. Boundaries cover Saskatoon HTL facility and injection operations.	Isometric Bio-Oil Geological Storage Protocol v1.1, Sections 1.2–1.3	Yes	Scope clearly defined for facility and storage site. Transport and ancillary emissions included.
Additionality	Project is financially additional: revenues from credits are required for commercial viability. Regulatory additionality confirmed: no laws mandate HTL or geological storage of bio-oil. Common practice additionality supported: few/no comparable operations in Canada. TRL internally	Standard Section 2.5.3	Yes	NuLife does not generate any revenue from the sale of bio-oil and is not financially viable without the generation and sale of carbon credits. There is no current mandate by the province or the country that enables to

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	assessed at 8; benchmarked against U.S. DOE (to be confirmed).			production of bio-oil from waste biomass
Uncertainty	Monitoring plan includes Option B sampling: bio-oil sampled ~3x daily, duplicates tested. Carbon wt% verified by elemental analysis at University of Saskatchewan. Brine tested monthly; continuous flow meters for injection volumes.	Standard Section 2.5.8	Yes	Monitoring plan robust, but external lab not ISO 17025 accredited. NuLife uses a lab that has provided the necessary details Isometric requested for non-ISO 17025 labs.
Permanence	Salt caverns in Prairie Evaporite Formation; regulated under CSA Z341.2. Caverns pressurized at ~300 psi; mechanical integrity testing every 5 years. Storage security >1,000 years. Pressure monitoring continuous; no negative pressure events recorded.	Protocol Section 2.5.8	Yes	Permanence supported by geological evidence and regulatory oversight. Additional documentation on potash seams/leakage history recommended.
Leakage (market effects)	Feedstocks are local waste residues (e.g., oat hulls, dairy DAF sludge, grain milling residues, FOG and oat hulls) that are not bought/sold as commodities and have no alternative energy/materials market; project does	Standard Section 2.5.4 Feedstock accounting module	Yes	No carbon market leakage expected because inputs are non-market wastes; crediting excludes any biomass that would remain

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	not alter supply or demand elsewhere. Supplier confirmations of waste status and no double-claiming: [redacted] documentation certifying that feedstocks are ineligible for energy use per protocol.			durably stored or be used for energy. Physical risks are covered under Permanence/Risk of Reversal, not Leakage.
Ownership	NuLife holds operational control of HTL plant. [redacted] owns and operates storage caverns (agreement in place April 2025). Letters/contracts needed from [redacted] confirming no other claims on environmental attributes.	Protocol Section 3.1	Yes	Ownership arrangements have been shared.

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Appendix

Appendix 1: List of Issues

#	Type	Date Issued	Issue Description	Resolution	Impact on material misstatement? (Y/N)	Impact on non-conformance? (Y/N)	Impact on adverse verification statement if not addressed? (Y/N)	Date Resolved
1	Misstatement	21-Jul-2025	The value 0.084 only appears once in the GLEC Framework and refers to $kgCO_2e/t{\cdot}km$, not $gCO_2e/t{\cdot}km$. The referenced GLEC table also applies to Europe and South America, not North America.	NuLife clarified that the correct source is Page 98 for North American Road Transport. WtW values are listed in grams and converted to kilograms. Certify uses kilograms for transport. The unit label was corrected to $kgCO_2e/t{\cdot}km$. RESOLVED.	N	Y	N	04-Aug-2025

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2	Clarification	21-Jul-2025	The specific emission factor (2.01 $kgCO_2e/kg$) used in Certify for HTL site steel could not be located in documentation.	NuLife confirmed the EF originates from the Sphera database, which requires a paid license. Supporting screenshots from Sphera were uploaded to Certify and match the reported EF. RESOLVED.	N	N	N	24-Jul-2025
3	Clarification	21-Jul-2025	Residual mix terminology in electricity EF reporting was unclear in the operations template.	NuLife confirmed that Isometric approved renaming the input to 'Grid Emission Factor'. The residual mix reference remains acceptable under current module guidance. RESOLVED.	N	N	N	05-Aug-2025

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4	Clarification	22-Jul-2025	Eco EF details for concrete were unclear; reported EF (145.42 $kgCO_2e/m^3$) inconsistent with database values.	NuLife updated the EF to 358 $kgCO_2e/m^3$ as per Eco v3.11 cut-off method. New Excel LCA file uploaded with corrected value. RESOLVED.	N	N	N	05-Aug-2025
5	Clarification	24-Jul-2025	IRR value for 2023 appeared unusually large [redacted]	NuLife confirmed this reflects initial plant investment. Reasonable explanation accepted. RESOLVED.	N	N	N	05-Aug-2025
6	Clarification	24-Jul-2025	Methane emissions excluded in PDD avoided emissions baseline.	NuLife explained methane avoidance is excluded per Isometric guidance, since avoided emissions are not removals. RESOLVED.	Y	Y	N	05-Aug-2025

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7	Clarification	24-Jul-2025	Leakage table explanation needed; issue of feedstock costs not addressed clearly.	NuLife provided documentation (packing slips, waste acceptance certificates) confirming feedstock is a waste by-product with $0 cost. This satisfies Isometric leakage criteria. RESOLVED.	N	Y	N	05-Aug-2025
8	Clarification	24-Jul-2025	No selection marked in leakage table for oat hulls feedstock.	NuLife clarified oat hulls are an industrial by-product, not agricultural biomass, and Isometric advised compliance via EC6 and EC11. Updated table provided. RESOLVED.	N	N	N	05-Aug-2025

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9	Omission	05-Aug-2025	Electricity consumption input missing in Certify low-carbon procurement section.	NuLife deleted the unused input to resolve inconsistency. RESOLVED.	N	N	N	20-Aug-2025
10	Clarification	05-Aug-2025	Discrepancy between Certify (611 kWh) and LCA (651 kWh) electricity use.	NuLife clarified Certify uses power meter data (611 kWh); additional manual logs and equipment use account for the difference. Breakdown provided. RESOLVED.	Y	N	Y	20-Aug-2025
11	Clarification	05-Aug-2025	Evidence for horsepower (HP) and kW inputs for injection pump electricity use requested.	NuLife provided manufacturer pump manual and Excel file with calculations. Values verified as accurate. RESOLVED.	N	N	N	20-Aug-2025

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12	Recommendation	05-Aug-2025	Error in LCA file formula for HTL site steel embodied emissions.	NuLife explained formula accounts for end-of-life emissions grouping. No material misstatement, but update recommended for clarity.	Y	N	Y	20-Aug-2025
13	Forward Action Request	05-Aug-2025	Minor inconsistency in stainless steel EF between Certify (3.56) and Excel file (3.46).	NuLife updated Excel to match Certify EF of 3.56. RESOLVED.	N	N	N	20-Aug-2025
14	Clarification	05-Aug-2025	Additional LCA Excel requested for operations emissions to match project emissions reporting.	NuLife provided consolidated Excel with trial data and flow chart showing data pathway into Certify. Clarification accepted. RESOLVED.	N	N	N	20-Aug-2025

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15	Missing Requirement	19-Aug-2025	TRL analysis not included in PDD for common practice additionality demonstration.	NuLife added TRL analysis to PDD Section B.7.4. RESOLVED.	N	Y	Y	28-Aug-2025
16	Missing Requirement	19-Aug-2025	Injection site drilling permits not provided.	NuLife uploaded drilling permits for [redacted] injection site. RESOLVED.	N	Y	Y	28-Aug-2025
17	Missing Requirement	19-Aug-2025	Calibration records for truck scales missing.	NuLife provided calibration certificates in shared drive. RESOLVED.	N	Y	Y	28-Aug-2025
18	Missing Requirement	19-Aug-2025	Evidence of brine testing and analysis missing.	NuLife provided brine testing documentation and analysis. RESOLVED.	N	Y	Y	28-Aug-2025
19	Missing Requirement	28-Aug-2025	Evidence of workplace safety incidents requested.	NuLife confirmed no lost-time accidents since project start. The injection site has recorded [redacted]	N	N	N	28-Aug-2025

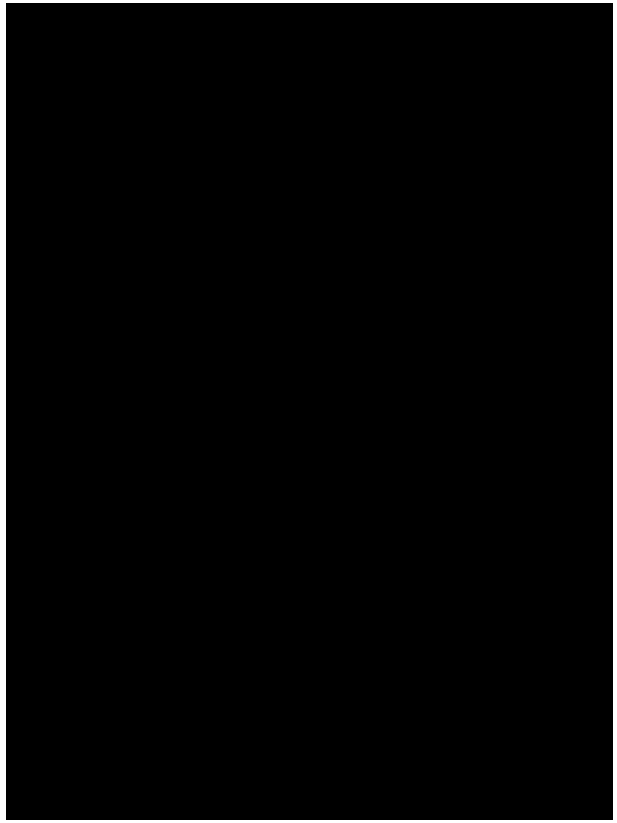
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				[redacted] RESOLVED.				

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Appendix 2. Pictures from site visit

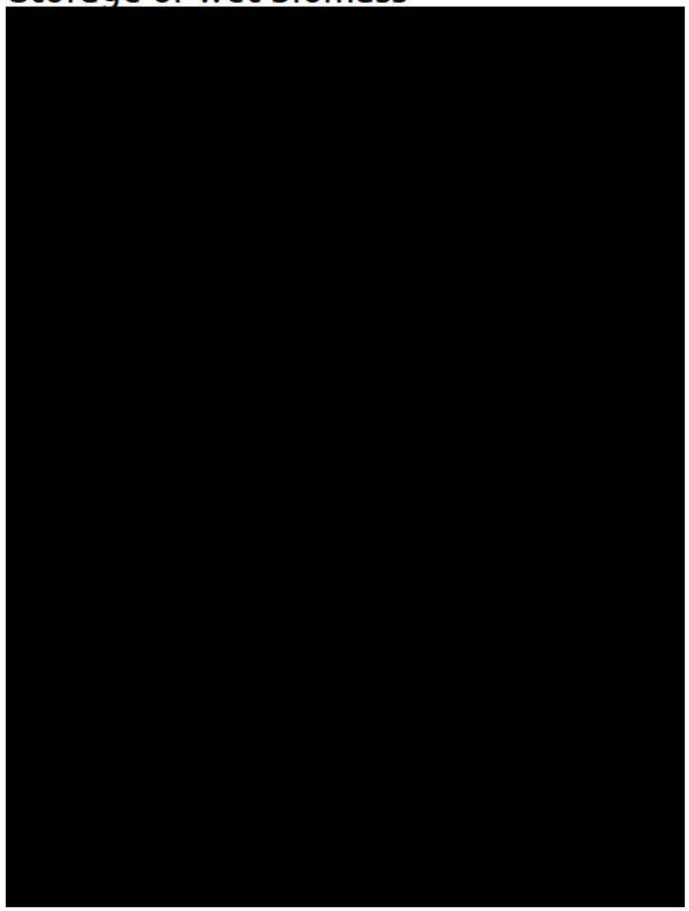
Storage tank of feedstock



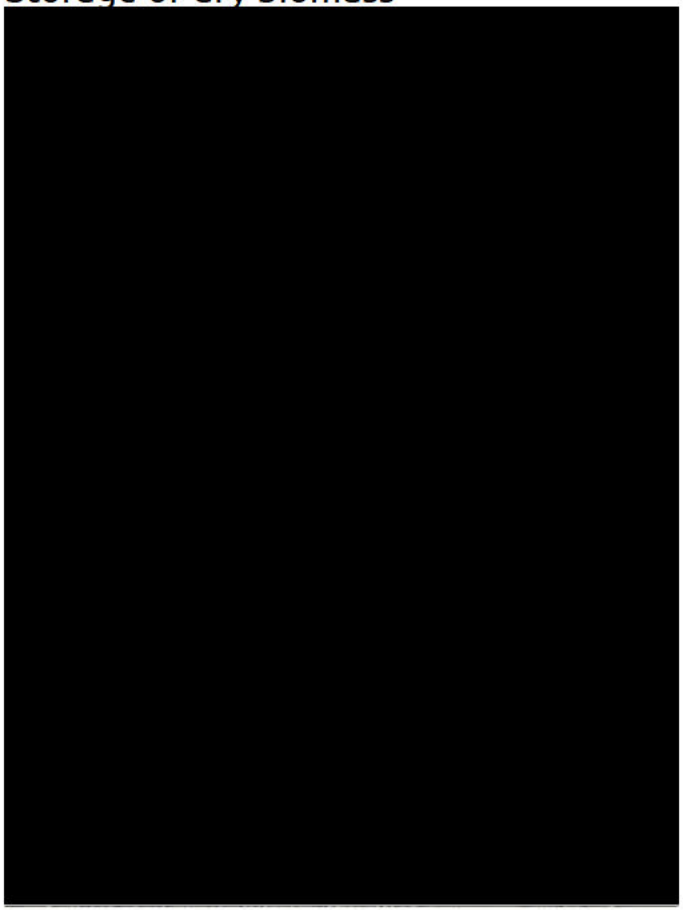
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Storage of wet biomass

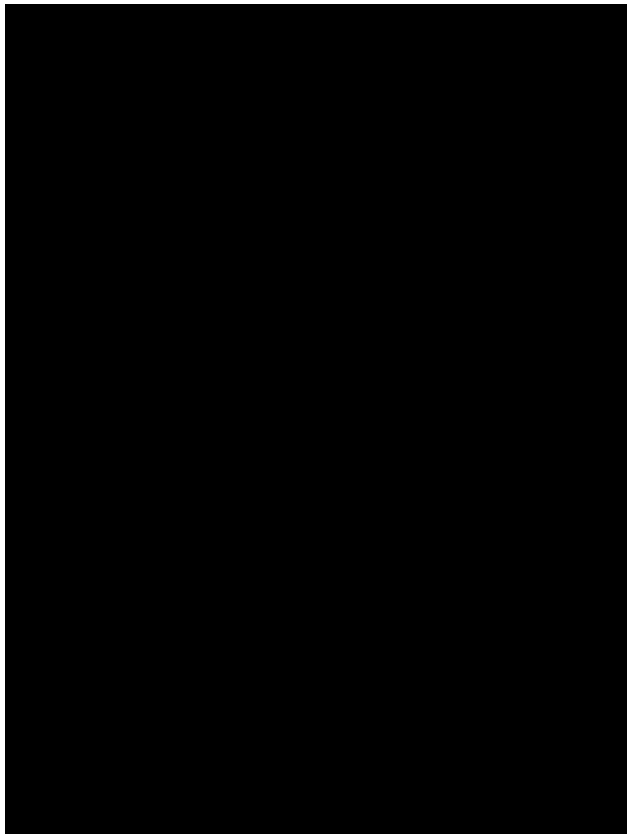


Storage of dry biomass



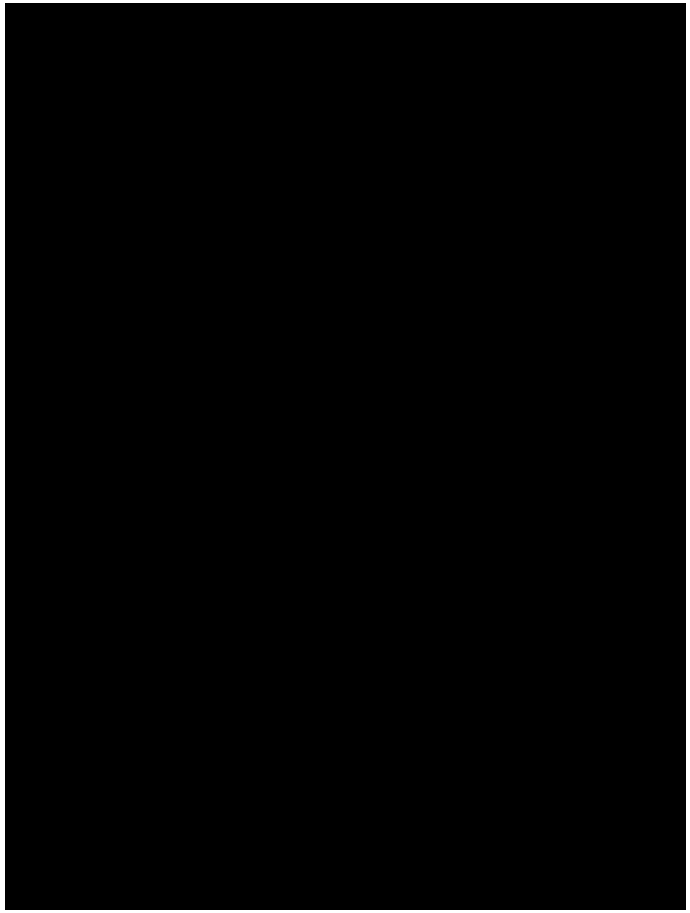
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Hydrothermal liquefaction processors (liquid biomass flows back and forwards through the pipes)

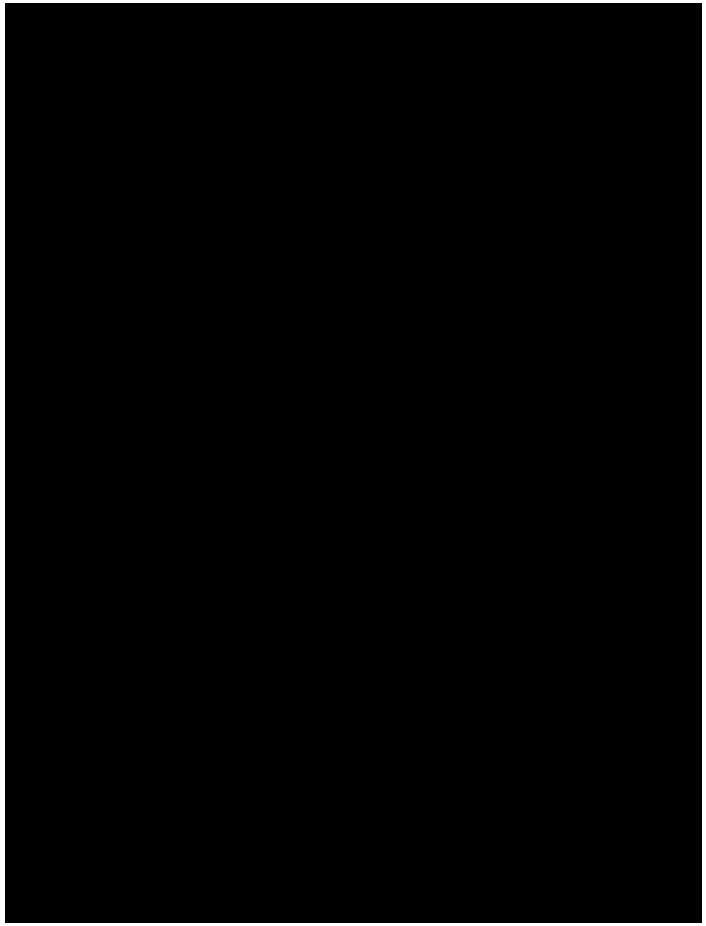


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Flow diagram of NuLife’s inputs and outputs.



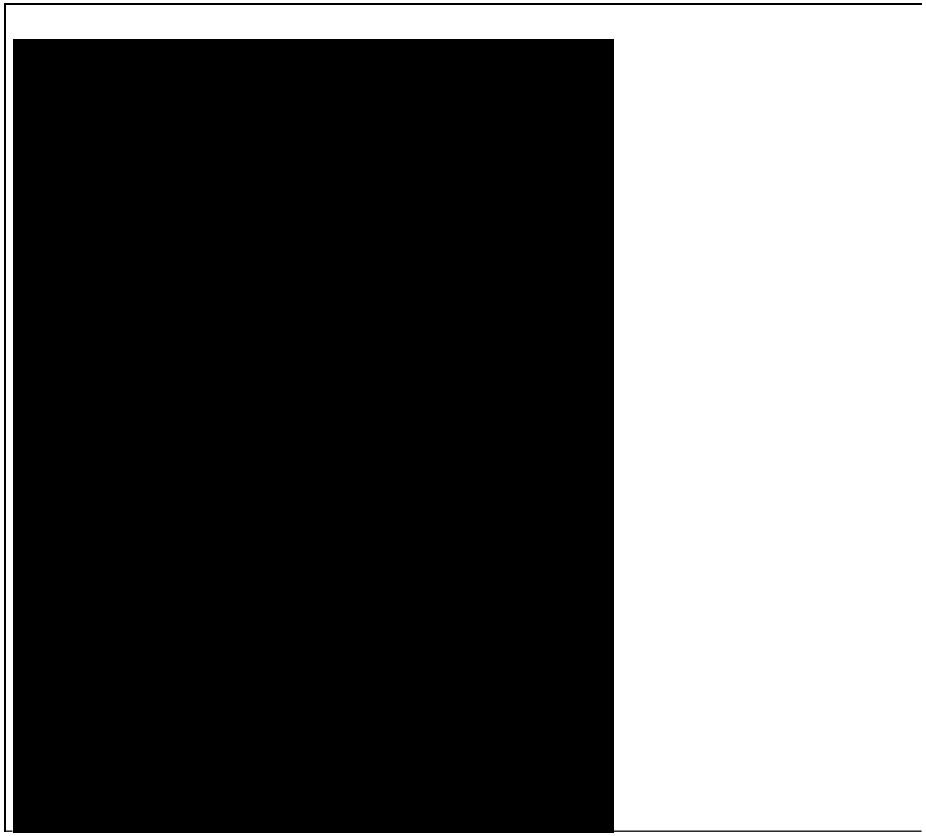
Example of tracking of cumulative energy consumed over given period.

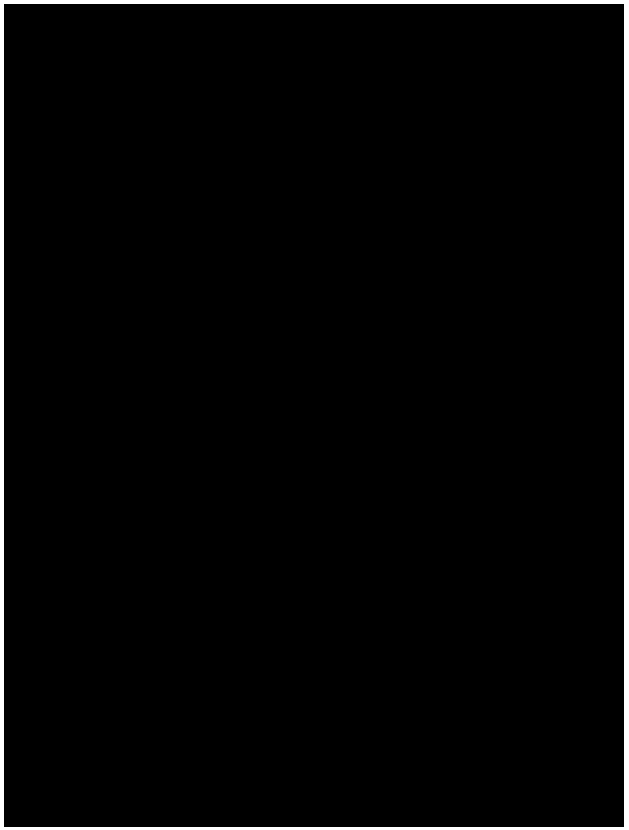


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Main power meter





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Appendix 3: Validation plan

Introduction

This validation plan outlines key areas to audit to ensure the project adheres to the Isometric Standard and the Bio Oil Geological Storage Protocol.

1. Project Eligibility and Protocol Adherence

- **Review Eligibility Criteria:** Verify that the project meets all applicability requirements outlined in Section 4.0 of the Bio-oil Geological Storage protocol. This includes feedstock eligibility, technology (HTL), injection method, net negativity, and environmental/social impact assessments.
- **Protocol Version:** Confirm that the latest available version of the "Bio Oil Geological Storage" Protocol (V1.1) is being used, or that any grace period for an earlier version is properly documented.

2. Project Boundary and Emissions Accounting

- **Boundary Confirmation:** Validate the defined project boundary, ensuring all relevant Greenhouse Gas (GHG) Sources, Sinks, and Reservoirs (SSRs) are included or justifiably excluded. Verify that the project follows Isometric's GHG Accounting Framework.
- **GHG Accounting:**
    - Verify calculations for included emission sources such as fuel use for transport (feedstock, bio-oil, wastewater, consumables), electricity use for HTL conversion and injection, and embodied emissions for equipment and consumables.
    - Review the justification for excluded sources like upstream biomass cultivation inputs, HTL tailgas biogenic CO2 emissions, and embodied emissions from the construction of existing facilities.
    - Assess the counterfactual baseline emissions, particularly regarding replacement emissions and temporary carbon storage from feedstock.

3. Leakage Assessment

- **Market Leakage:** Verify the assessment that market leakage associated with feedstock displacement is minimized due to the focus on waste streams.
- **Replacement Emissions:** Confirm that replacement emissions are determined in compliance with Section 3.2 of the Biomass Feedstock Accounting Module, and any quantification of leakage is appropriately performed.

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4. Additionality Demonstration

- **Financial Additionality:** Examine project financials, including the Internal Rate of Return (IRR) analysis and sensitivity analysis, to confirm that the project is not financially viable without carbon credit sales.
- **Environmental Additionality:** Validate the net negative carbon removal claim by assessing the Net Negativity work on Isometric's Certify Platform.
- **Regulatory Additionality:** Confirm that the project is not mandated by any existing regulatory, policy, or legal requirements.

5. Durability Assessment and Monitoring

- **Durability Confirmation:** Verify the justification for the 1,000+ year durability of the carbon sequestration in salt caverns, including references to scientific literature and site-specific information.
- **Monitoring Plan:**
    - Review the submitted Monitoring Plan and compare it against the "Monitoring Plan Requirements" appendix of the Bio-Oil Geologic Storage module.
    - **Bio-Oil Carbon Content:** Verify the methodology for determining the carbon weight percentage of the bio-oil injectant (e.g., ASTM D5921 or equivalent, minimum 3 samples per batch, duplicate testing, outlier handling). Confirm that the University of Saskatchewan Chemical Engineering Lab, an in-house lab, provides additional required information to Isometric for acceptability.
    - **Bio-Oil Mass:** Validate the method for determining the total mass of bio-oil injectant (density measurement with moisture deduction, ASTM D4052, ASTM E-1756).
    - **Tailgas Monitoring:** Confirm the measurement of mass flow rate and concentration of CO2 and CH4 in tailgas via GC Analysis.
    - **Electricity Usage:** Verify the measurement of electricity consumption (kWh) per HTL production run using a power meter and manual records for non-metered equipment. Confirm purchase and retirement of RECs.
    - **Transport Emissions:** Review the Mass-Distance-Based method for calculating transport emissions for feedstock, bio-oil, consumables, wastewater, and staff travel. Verify data sources (Bill of Lading, scale tickets, supplier invoices, Google Maps, vehicle reimbursement logs).
    - **Embodied Emissions:** Confirm the mass-based method for calculating embodied emissions from HTL conversion and injection site equipment, allocated over the project lifetime.
    - **Consumables Emissions:** Verify the methods for accounting for emissions from municipal washing water, hydrogen peroxide, sodium hydroxide, and activated carbon.
    - **Brine and Gas Testing (Injection Site):**
        - Review pre-injection (baseline) brine testing for pH, conductivity/chloride, temperature, and carbon content.
        - Review pre-injection (baseline) gas testing from the brine tank head space.
        - Verify ongoing injectant monitoring for pH, temperature, and chloride content.
        - Assess displaced brine testing frequency and parameters (carbon wt%, temperature, pH, chloride content).
        - Confirm ongoing gas testing using a fixed gas monitor and subsequent GC/IRMS testing in case of alarms to detect reversals.

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- **Cavern Integrity:** Review documentation related to cavern integrity, including Mechanical Integrity Testing (MIT) (pressure testing, casing scanning), corrosion testing, cavern pressure, volume (sonar survey), and USDW quality monitoring, all in accordance with CSA Z341.2 and [redacted]

6. Data Collection and Storage

- **Data Handling:** Confirm procedures for data transmission, collection, and storage (Excel, Isometric's Certify platform, Nulife internal software, etc).
- **Record Archiving:** Verify the length of time records are archived and the backup procedures in place.
- **Responsibility:** Identify the person(s) or organization(s) responsible for measurement and data collection.

7. Environmental and Social Impacts

- **Impact Assessment:** Review the assessment of environmental and social impacts and co-benefits, including resource efficiency, pollution prevention, waste generation, and hazardous material release.
- **Mitigation Plans:** If applicable, verify the mitigation plans in place to prevent identified risks.
- **Grievance Mechanism:** Review the mechanism for stakeholders to voice concerns.

-END OF REPORT-

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